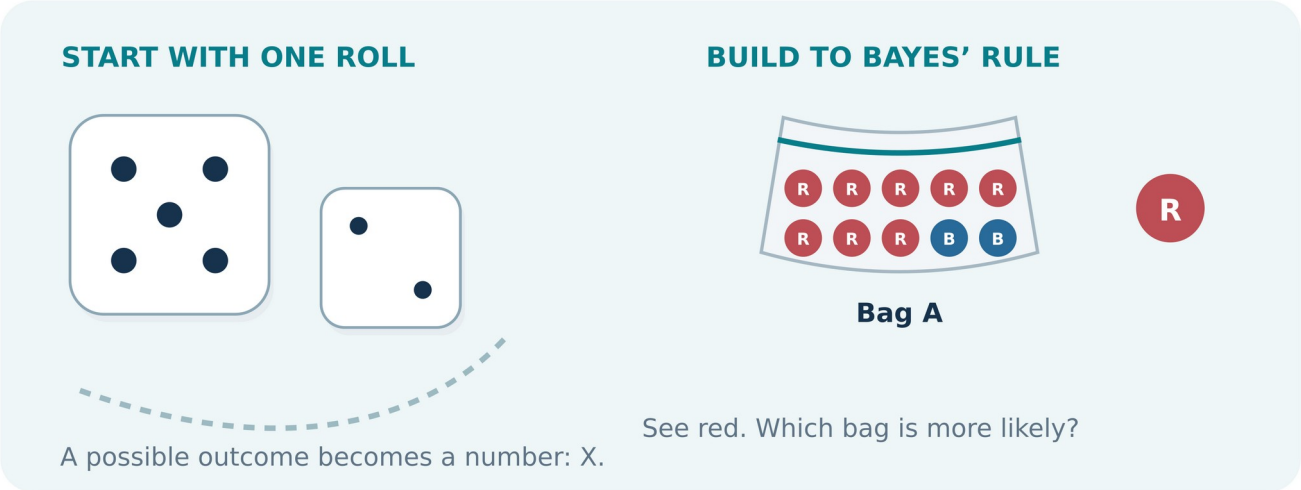
WEEK 09 / CS203 / PRACTICAL LAB

Probability experiments and Bayes' rule

From one die roll to reasoning from evidence.



New Uzbekistan University · Data Science Fundamentals

Mohammad Sabokrou

How to work through this lab

READ THE EXPLANATION · CALCULATE BY HAND · WRITE CODE · EXPLAIN

For each part, read the explanation, calculate the answer by hand, write your own program, then explain what the results mean. The same experiments develop gradually from a fair die to Bayes' rule. You can complete this lab before or after the lecture.

Write ordinary programs in your usual Python editor. Keep your code, hand calculations, result tables and short explanations. Use the course's existing in-class review and submission process. The activities use mathematical models and fictional bag experiments.

Your route through the lab

Part	Idea	Time
01	Probability and a random variable	10 min
02	What happens after ten thousand rolls	15 min
03	The distribution of the random variable	10 min
04	Conditional probability means among	15 min
05	An experiment with two bags	15 min
06	Bayes' rule answers the reverse question	20 min
07	A different prior changes the answer	15 min

Plan for 2 hours, including a 10 minute break after Part 4 and a final discussion.

PART 01 / 10 MINUTES

Probability and a random variable



Probability describes how likely an event is. A random experiment has an uncertain result. Rolling a die is a random experiment. One outcome is the number on its upper face. An event is a description that some outcomes satisfy, such as “the number is greater than 4”.

Let X be the number shown by one roll. X is a random variable: a rule that assigns a number to each outcome. Before rolling, we do not know its value. After rolling, we observe one value, such as X = 5. The possible values are 1, 2, 3, 4, 5 and 6.

For a fair die, these six outcomes are equally likely. The probability of an event is its number of matching outcomes divided by 6. P(X > 4) means “the probability that X is greater than 4”. A probability of 0 means impossible, and 1 means certain.

Example: only one face shows 1, so P(X = 1) = 1/6. To answer another event, first list the faces that match it.

Calculate by hand

1. List the outcomes satisfying X > 4. Calculate P(X > 4) as a fraction and a decimal.

2. Calculate P(X = 4) and P(X ≤ 4). Check that P(X > 4) + P(X ≤ 4) = 1.

Write and run your program

1. Write a short program containing the six possible die values.

2. Count how many values satisfy X > 4, then divide by the total number of possible values. Print the probability.

3. Modify the condition to calculate P(X = 4) and P(X ≤ 4). Compare with your hand calculations.

Explain what you observe

1. What does X represent? What is the difference between the event X > 4 and one observed value such as X = 5?

Keep Your hand calculations, the program, and its three exact probabilities.

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PART 02 / 15 MINUTES

What happens after ten thousand rolls



One roll

10,000 rolls

Generate • store • count • compare

An exact probability comes from the mathematical model. A simulation uses a computer to repeat that experiment. Here each simulated roll must choose one of the integers 1 to 6 with equal chance. We assume independent rolls: one roll does not change the chances on the next roll.

After N rolls, an observed relative frequency is the number of rolls satisfying the event divided by N. For example, if an event occurs 7 times in 20 trials, its observed relative frequency is 7/20. This is an estimate of the probability, based on the simulated outcomes.

With many independent rolls of a fair die, relative frequency tends towards the exact probability. It need not equal that probability in a finite run, and it can temporarily move farther away as more rolls are added.

To predict an approximate event count, multiply N by the exact probability. For an event with probability 1/2, 1000 trials give an expected count of 500. The actual count can differ.

Calculate by hand

1. Using your answer from Part 1, predict approximately how many of 10000 rolls will have X > 4.

2. Must that count occur exactly? Explain your prediction before running the program.

Write and run your program

1. Write a program that generates and stores 10000 fair die rolls in a list. Keep this list for Parts 3 and 4.

2. For N = 10, 100, 1000 and 10000, inspect the first N stored rolls. Count X > 4 and calculate count/N. Print the results in a table.

3. Generate a new list of 10000 rolls and repeat once. Record both large-run results.

Small programming hints

`random.randint(1, 6)` After import random, this generates one integer from 1 to 6, including both endpoints.

`for i in range(N)` A loop can repeat your roll-generation step N times.

Explain what you observe

1. Compare the simulated estimates with your exact probability. Why can two runs give different answers?

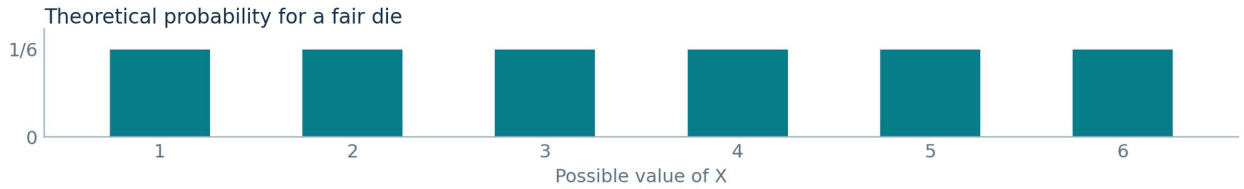
2. Did the estimate become closer at every listed N in your run? Explain why this is not guaranteed.

Keep A table with N, count of X > 4 and count/N, plus the second 10000-roll result. Use the original list in Parts 3 and 4.

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PART 03 / 10 MINUTES

The distribution of the random variable



A distribution describes the possible values and how much probability belongs to each value. The random variable X tells us what number we record. Its distribution tells us how likely the possible numbers are.

For the fair die, X takes values 1 to 6. Each has theoretical probability 1/6. In your simulated data, the observed frequency of a face is its count. Divide that count by the number of rolls to obtain its observed proportion.

A bar chart can show this distribution. Put the face values on the horizontal axis and the observed proportions on the vertical axis. Equal theoretical probabilities do not force exactly equal counts in a finite simulation.

> If face 2 appears 1650 times in 10000 rolls, its observed proportion is 1650/10000 = 0.165. The theoretical probability remains 1/6.

Calculate by hand

1. Write the theoretical probability distribution for X = 1, 2, 3, 4, 5 and 6. Check that its probabilities sum to 1.

2. Predict the approximate count for each face in 10000 rolls. Must all six counts be identical?

Write and run your program

1. Using the original stored list, count each face from 1 to 6.

2. Print a table with face, observed count, observed proportion and theoretical probability.

3. Draw a labelled bar chart of the six observed proportions. Check that the counts total 10000 and the proportions sum to 1.

Explain what you observe

1. Which face had the largest observed proportion? Does that single result prove the die is unfair?

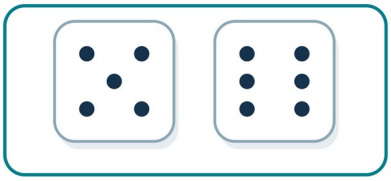
2. Explain the difference between one observed value of X and the distribution of X.

> Keep A six-row frequency table and a labelled probability bar chart.

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PART 04 / 15 MINUTES

Conditional probability means among



Given X > 4

Only these outcomes remain.

New information can restrict the outcomes we consider. Conditional probability asks for a probability within that restricted group. Read P(A | B) as “the probability of A, given B”, or “among the outcomes satisfying B, what fraction satisfy A?”

The denominator is the size of the group after the vertical bar. The numerator counts outcomes that satisfy both conditions. For equally likely outcomes, P(A | B) = count satisfying both A and B / count satisfying B. The denominator must be positive.

Reversing the condition asks a different question. P(A | B) and P(B | A) can therefore have different answers.

Suppose you know X > 4. Only 5 and 6 remain possible. One of these two values is 6, so P(X = 6 | X > 4) = 1/2.

Calculate by hand

1. Let A mean X > 4 and B mean X ≥ 4. List the faces in A and the faces in B.

2. Calculate P(A | B) and P(B | A). State the denominator for each.

Write and run your program

1. Use the original 10000 die rolls. Count rolls in A, rolls in B, and rolls satisfying both.

2. Estimate P(A | B) by dividing the both-count by the B-count. Estimate P(B | A) by dividing the both-count by the A-count.

3. Print both estimates with their counts and denominators. Compare them with your hand calculations.

Explain what you observe

1. Why should you not divide by 10000 for these conditional questions?

2. Explain the two answers using sentences starting with “Among…”.

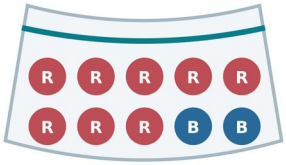
Keep Two exact calculations and two estimates, with an explicit numerator and denominator for each.

Take a 10 minute break after this part.

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PART 05 / 15 MINUTES

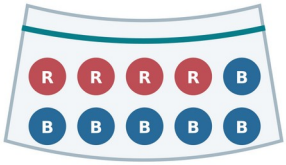
An experiment with two bags



Bag A: 8 red, 2 blue

Choose a bag, then one ball.

Return it after every trial.



Bag B: 4 red, 6 blue

Bag A contains 8 red balls and 2 blue balls. Bag B contains 4 red balls and 6 blue balls. In each trial, first choose a bag with equal chance, then choose one ball uniformly from that bag. Return the ball before the next trial, and choose a bag again in every trial.

Let A mean “Bag A was chosen”, B mean “Bag B was chosen”, and R mean “the ball is red”. P(R | A) is the chance of red when we already know the bag is A. P(A and R) is the chance that the trial both chooses A and produces red.

Think through the routes. Multiply the probability of choosing a bag by the red probability inside it. Add the red results from the two routes to obtain the overall probability of red. The routes do not overlap because each trial chooses only one bag.

> Bag A has 2 blue balls out of 10, so P(blue | A) = 2/10. To find an expected count for a route, take the expected number entering that bag and multiply by its colour proportion.

Calculate by hand

1. Calculate P(A), P(B), P(R | A) and P(R | B).

2. In 1000 trials, the expected number choosing each bag is 500. Complete the table of expected red and blue counts.

3. Use the table to calculate P(A and R), P(B and R) and P(R). These are theoretical expectations, not guaranteed counts in a simulation.

Chosen bag	Expected red	Expected blue	Expected total
A	Calculate	Calculate	500
B	Calculate	Calculate	500
Total	Calculate	Calculate	1000

Write and run your program

1. Simulate 10000 trials. In each trial, choose A or B with equal chance, then make a fresh random selection of a ball from that chosen bag. Store the bag and colour.

2. Print the observed two-way count table: Bag A or B versus red or blue.

3. From your counts, estimate P(R | A), P(R | B) and the overall P(R). Compare with your mathematical answers.

Small programming hints

`random.random()` Returns a number from 0 inclusive to 1 exclusive. A value below 0.5 can select A; otherwise select B.

`A new random.randint(1, 10)` Can choose a ball position. Positions 1–8 are red in A; positions 1–4 are red in B.

Explain what you observe

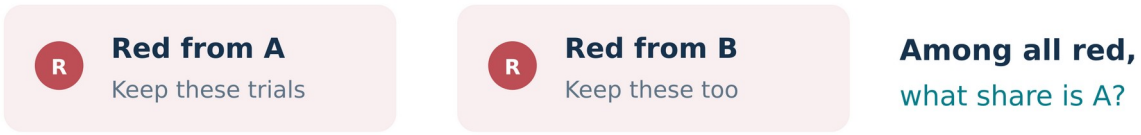
1. What is one trial in this experiment? Why do we return the ball and choose a bag again?

2. Why does the overall probability of red include results from both bags?

> Keep The completed expected-count table, an observed count table, and three probability estimates.

PART 06 / 20 MINUTES

Bayes' rule answers the reverse question



Suppose the experiment has already produced a red ball, but you did not see which bag was chosen. Now ask: "Among red outcomes, what fraction came from Bag A?" This is P(A | R). It reverses the direction of P(R | A).

Use your expected-count table first. Keep only the red column. Divide the expected red count from A by the total expected red count from both bags. This is the same conditional-probability idea used for the die.

Bayes' rule expresses that count argument as a formula: P(A | R) = P(R | A)P(A) / P(R). Here P(R) = P(R | A)P(A) + P(R | B)P(B). The prior P(A) describes the chance of choosing A before seeing the colour. The posterior P(A | R) describes that chance after learning that the ball is red.

If a separate example contains 30 red outcomes from A and 20 red outcomes from B, then among the 50 red outcomes, the fraction from A is 30/(30 + 20). Both routes belong in the denominator.

Calculate by hand

1. Using the 1000-trial expected table from Part 5, calculate P(A | R). Show the red count from A and the total red count.
2. Calculate the same answer with Bayes' formula. Substitute each probability before simplifying.
3. Compare P(A | R) with P(R | A). Do they describe the same group?

Write and run your program

1. Use your simulated records from Part 5. Keep only trials where the ball is red.
2. Calculate the fraction of those red trials that chose Bag A. Print the numerator and denominator.
3. Write a few lines that also calculate the exact Bayes result from the model probabilities. Display the exact answer beside the simulated estimate.

Explain what you observe

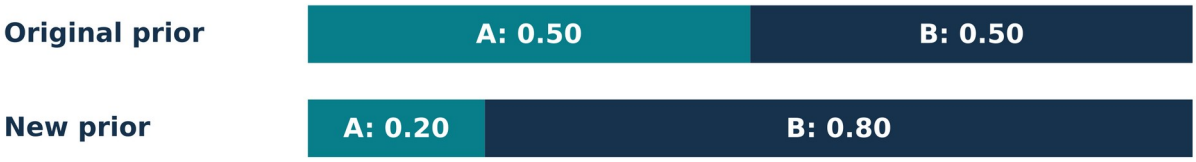
1. Explain why the denominator includes red balls from Bag B even though the question asks about Bag A.
2. Did seeing red change the probability of Bag A compared with its prior? Explain using your mathematical result.

Keep The count-based and formula-based calculations, the simulated inverse probability, and a short explanation.

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PART 07 / 15 MINUTES

A different prior changes the answer



The bag contents stay exactly the same. Now the selection mechanism chooses Bag A with probability 0.20 and Bag B with probability 0.80. Only the prior changes. P(R | A) and P(R | B) remain determined by the same bag contents.

A bag that is chosen less often contributes fewer outcomes. To interpret evidence about the colour, we must account for how often each bag was selected as well as how red its contents are.

The model probabilities are exact under the stated assumptions. A larger simulation helps estimate their consequences more steadily. It does not change the exact Bayes answer.

In 1000 trials with P(A) = 0.20, the expected number choosing A is 1000 × 0.20 = 200. The remaining 800 choose B. Apply each bag's red proportion within its own group.

Calculate by hand

1. Calculate expected red counts from A and B among 1000 trials with the new prior.

2. Calculate the overall P(R) and then P(A | R). Compare with Part 6.

3. Before coding, predict whether the posterior will rise or fall when P(A) changes from 0.50 to 0.20. Give a reason.

Write and run your program

1. Modify the bag-selection step so a random number below 0.20 chooses A. Keep the ball-selection rules unchanged. Simulate 10000 trials.

2. Estimate P(R) and P(A | R) from the observed counts.

3. Create a comparison table for priors 0.50 and 0.20. Include the exact and simulated P(R) and P(A | R).

Explain what you observe

1. Why can the same red ball evidence lead to different posterior probabilities when the prior changes?

2. In 3–4 sentences, explain probability, a random variable, simulation and Bayes' rule using the die and bag experiments.

Keep A two-row comparison table, the code you wrote, and a short final explanation.

Final check

Show how your code calculates each probability. Compare exact answers with estimates. In every conditional probability, identify the group in the denominator.

Reading: OpenIntro Statistics, 4th edition, sections 3.1, 3.2 and introductory 3.4. www.openintro.org/book/os/

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